

Master Thesis Pre-Analysis Plan

LLM in Court: Leveraging Open-Source Language Models to Enhance Empirical Legal Research in Brazil

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1. Summary

This study aims to harness open-source language models to advance empirical and quantitative research in Brazilian courts and other Portuguese-speaking jurisdictions. The dataset—already collected through publicly available repositories and direct court websites—contains a broad range of judicial decisions, primarily from the São Paulo Court of Justice. The central objective is to examine how these models can effectively perform information extraction, entity recognition, and summarization tasks, while maintaining high accuracy, computational efficiency, and the ability to navigate the complexities of legal language.

Leveraging previously conducted empirical research (as a strategy for resource and emission savings), we will use datasets already annotated by experts. Decisions related to the cases used will be collected via web scraping, and new decisions will be gathered for subsequent model application. Following this, a selection of open-source language models—including LLaMA, LOLA, Mistral, GPT-2, Phi, Gemma, and DeepSearch—will be evaluated and fine-tuned within a consistent prompt framework tailored to legal analysis. Baseline performance metrics will guide potential optimization techniques, such as pruning or distillation, aimed at reducing computational overhead.

Finally, both the curated dataset and a cost-effective fine-tuned model will be made available on platforms such as HuggingFace, enabling large-scale, data-driven analyses of legal texts. By broadening access to empirical legal tools, this initiative seeks to provide a stronger, evidence-based foundation for public policy and legal research, thereby reinforcing the quantitative dimension of legal studies.

2. Motivation and background

Empirical legal research is essential for shaping effective public policies by ensuring that judicial and legislative decisions are grounded in rigorous, data-driven analysis rather than intuition or anecdote. In Brazil and many other jurisdictions, legal scholarship has traditionally relied on qualitative and doctrinal approaches, which, while insightful, often fail to capture systemic biases or the real-world impact of legal decisions. This gap hinders the development of evidence-based policies that address social and judicial inequalities, especially in critical areas such as drug policy, criminal sentencing, and disparities in treatment based on gender and race.

The Brazilian legal landscape, characterized by underexplored issues like judicial bias and sentencing disparities, urgently needs empirical research to support policy reform. Without systematic data analysis, policymakers and legal practitioners lack the reliable evidence needed to craft interventions that effectively address injustices in the justice system. This shortfall not only limits the capacity for meaningful reform but also risks misdirecting policy efforts, thereby perpetuating inefficiencies and inequities.

Open-source language models and advanced computational legal research offer a transformative solution by enabling large-scale, systematic analysis of judicial trends and biases. By integrating machine learning, natural language processing, and empirical methods, this research will develop cost-effective, domain-specific models that provide deep insights into Brazilian court decisions. This approach not only strengthens academic inquiry but also equips policymakers and legal professionals with practical tools for fostering a more transparent and evidence-based legal system.

3. Introduction

The ability to use machine learning tools for extracting key information from legal documents has grown in strategic importance. Courts produce vast amounts of textual data that demand specialized, efficient, and transparent methods for annotation, classification, summarization, and analysis. Recent studies on large language models (LLMs) emphasize that simple off-the-shelf prompting is often insufficient for complex, domain-specific tasks, such as interpreting legal statutes or synthesizing intricate court rulings. Specialized fine-tuning offers a promising route to address these gaps in a cost-effective manner, enabling targeted legal tasks with relatively modest datasets.

Against this backdrop, a number of recent papers have explored domain-adapted LLMs, focusing on areas such as U.S. Supreme Court judgments, multi-jurisdictional summarization datasets, low-resource annotation strategies, specialized Portuguese LLMs, and local legal sentencing data. This section summarizes six central works that form the basis for the proposed research in Brazilian empirical legal studies, each subtopic highlighting their goals, data, methods, achievements, and relevance.

Dominguez-Olmedo et al. (2024) “Lawma: The Power of Specialization for Legal Tasks” evaluated whether fine-tuning an open-source LLaMA 3-based model on legal corpora could match or surpass GPT-4 for classification tasks covering the U.S. Supreme Court Database and Courts of Appeals. Their approach involved a consistent prompt framework with limited labeled data, showing double-digit accuracy gains over GPT-4. This underscores how specialized domain adaptation can outperform commercial black-box APIs, an insight especially pertinent to jurisdictions (like Brazil) seeking robust, open-source solutions to handle large volumes of judicial decisions. Also, they have showed that a relatively small amount of labeled data (about 100 cases) can yield strong results, which is particularly relevant for resource-constrained settings like Brazilian legal research.

Corrêa et al. (2024) “Tucano and GigaVerbo” focus on Portuguese NLP by releasing Tucano, a series of open-source language models, and GigaVerbo, a 200-billion-token corpus that spans journalism, academic texts, and more. By thoroughly documenting training code, data-cleaning protocols, and model checkpoints, they enable easy replication and fine-tuning for domain-specific tasks. These transparent workflows exemplify a scalable blueprint for model adaptation, highlighting the significance of reproducibility and language-specific tokenization for Portuguese texts. In terms of achievements, Tucano demonstrated substantial improvements over generic multilingual models for Portuguese, proving the value of custom tokenizers, large corpora, and publicly available checkpoints. This is highly relevant to Brazilian judicial contexts, where specialized legal language and local idiomatic usage require domain-adapted text representations. Tucano thus offers a robust starting point for fine-tuning on legal corpora, facilitating more accurate document classification, entity extraction, and summarization in low-resource language scenarios.

Junior et al. (2024) introduced Juru, a large language model pre-trained on 1.9 billion tokens from authoritative Brazilian legal sources—federal laws, academic papers, and court rulings. Focusing on classification tasks for multiple-choice law exams, Juru showed a notable boost in domain-specific performance, albeit with some trade-off in general-language tasks. Its main achievement is demonstrating that smaller, high-quality corpora can significantly boost legal reasoning. For Brazilian empirical studies, Juru highlights a direct route to develop domain knowledge while keeping computational footprints manageable.

Lacerda e Silva (2019) investigates disparities in drug-related sentencing in São Paulo, examining how gender, race, and age intersect to influence judicial decisions. In *Punindo as Diferenças*, Lacerda e Silva (2019) collected 352 drug-related sentencing decisions, systematically tabulating variables (race, gender, age, drug quantity, and so on) and applying jurimetric methods. The main achievement lies in uncovering potential extralegal biases—e.g., older Black men often receiving harsher sentences—offering a critical empirical window into structural discrimination. This highlights the urgency of domain-focused NLP tools: accurate extraction of sentencing patterns can lead to data-driven reforms, crucial for bridging racial and gender inequities in the Brazilian justice system. Also, this work is very important because it offers a high-quality dataset that can be used to train and test NLP models in the Brazilian legal context, specially in drug-related research.

It is also noteworthy: Santosh et al. (2024) introduced a specialized legal summarization benchmark, demonstrating that generative methods significantly enhance the clarity of judicial text summaries, which can condense multi-page opinions into concise briefs for efficient large-scale analysis. Similarly, Adhikary et al. (2024) showcased a weakly supervised strategy to expand small annotated datasets for attribute extraction, reducing the human annotation burden and improving classification accuracy—an approach particularly relevant for scaling analysis in Brazilian courts where manual labeling is impractical.

All six lines of inquiry converge on the value of specialized, domain-focused fine-tuning. They collectively argue that curated data and methodical adaptation substantially outperform generic prompting in legal tasks, even when only moderate-sized labeled sets are available. This lesson is critical for empirical legal studies in Brazil, where massive volumes of text—from criminal decisions to appellate rulings—remain largely untapped.

4. Research question

How can open-source language models be optimally deployed in the Brazilian legal context to accurately extract information, recognize entities, and summarize complex legal documents, considering factors such as cost, computational efficiency, and the specific challenges of legal language and document structure?

5. Data and Methods

This project integrates multiple publicly available and curated datasets of Brazilian judicial decisions to develop an empirically grounded approach. From the São Paulo Court (TJ-SP), I have 3 million decisions containing structured metadata (e.g., case ID, court, magistrate), all hosted on Google Cloud BigQuery and suitable for large-scale text processing. Additionally, I have acquired two high-quality labeled datasets: Benatti’s domestic violence dataset provides 160 expertly annotated rulings with insights into gender biases, while Lacerda e Silva’s (2019) drug sentencing corpus includes 300+ decisions rich in demographic and sentencing variables. Together, these sources ensure coverage across diverse legal domains, from family law to criminal drug offenses.

For methods, I will combine NLP pipelines and statistical analyses. First, I will clean and tokenize the texts focusing on removing boilerplate elements and capturing essential named entities (judges, parties, legislative references). Pretrained open-source LLMs (e.g., LLaMA 3, Mistral, Gemma) form the baseline. Each will be assessed zero-shot on extracting key attributes (like charges, defendant demographics) before fine-tuning.

Fine-tuning centers on verifying model accuracy against gold-standard references, particularly in attribute extraction and possibly classification and summarization if time allows. LoRA or

full fine-tuning will be employed on TPU/GPU infrastructure, balancing resource constraints with performance needs. Core metrics include accuracy, F1, and perplexity for generation tasks. If improvements are strong, I may compress the resulting models via pruning or distillation to lower computational demands.

A successful outcome involves producing an open-source, domain-tuned model on Hugging Face, plus a robust annotated dataset for replicable experimentation. Ideally, this model can auto-extract sentencing details and party attributes, generating concise legal summaries. Lastly, if data permits, I will assess how partial marijuana decriminalization (post mid-2024 STF ruling) shifted judicial sentencing, demonstrating the model’s capacity to track policy changes.

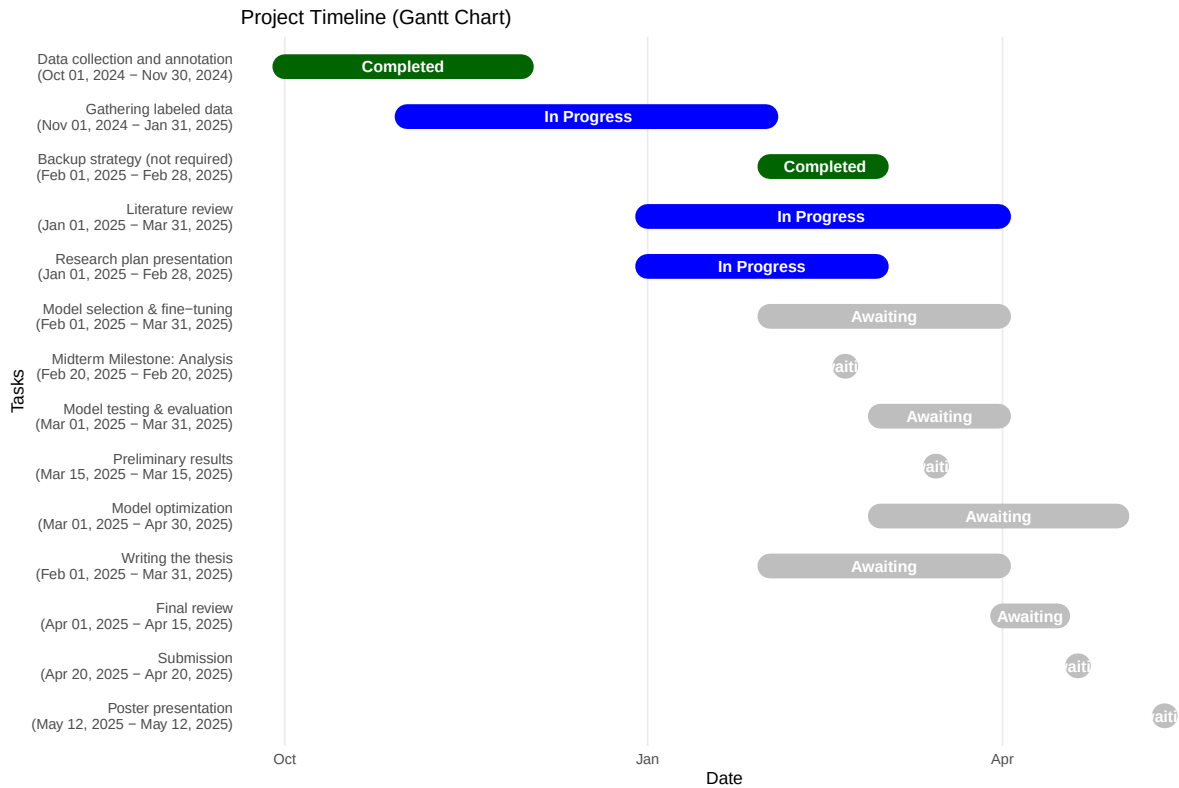
This streamlined approach—coupling curated legal corpora with targeted LLM fine-tuning—will deliver both practical analysis tools and a demonstration of how specialized NLP pipelines can inform policy by highlighting hidden biases or systemic trends in Brazilian court decisions.

6. Expected Results

By the conclusion of this research, the project is expected to yield a cost-effective, domain-tuned legal LLM that is specialized for the Brazilian judicial context and openly available on platforms such as Hugging Face. This model will be optimized to extract key information from judicial texts—ranging from sentencing details and party demographics to legal summarizations—using curated datasets of Brazilian court decisions. The high-quality, annotated corpora assembled (including the TJ-SP dataset, Benatti’s domestic violence cases, and Lacerda e Silva’s drug sentencing records) will enable reproducible evaluations and facilitate cross-jurisdiction comparisons, ultimately serving as a robust foundation for further legal AI research.

Additionally, the project aims to demonstrate the practical utility of the model in generating automated, data-backed legal reports. As a preliminary application, the model may be used to analyze the impact of the Supreme Federal Court’s June 2024 decision on partial marijuana decriminalization by extracting and summarizing sentencing trends and emergent biases. Overall, these achievements will contribute to a more transparent, empirical, and evidence-based framework for legal decision-making in Brazil, thereby informing policy discussions and reducing reliance on expensive commercial APIs or labor-intensive human annotation.

7. Timeline of the project



8. References

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